

Unleashing the Representational Power of Fourier Shapes for Attacking Infrared Object Detection



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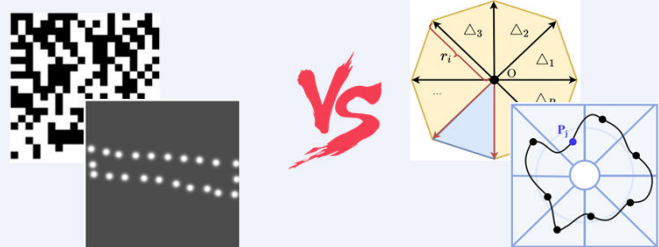
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Background & Motivation

Attack information carrier in infrared adversarial patch:

★ Internal grayscale texture

★ Geometric shapes



Texture-Based

Geometrically discrete

Difficult in physical deployment

Shape-Based

Poor shape representation

Undifferentiable

Core challenge:

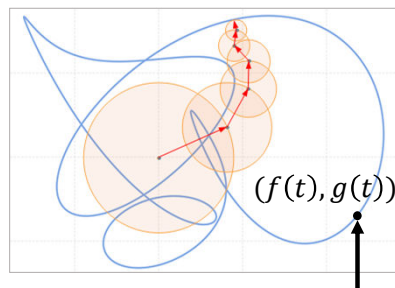
1. Powerful shape representation?
2. Gradient optimization efficiency?

Our solution:

1. Fourier shape representation
2. Differentiable shape-to-image mapping

Method Description

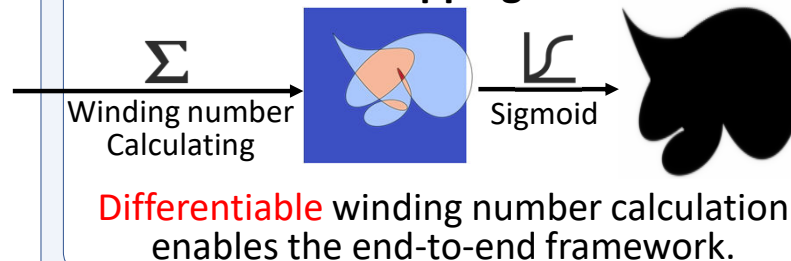
Fourier Shape Representation



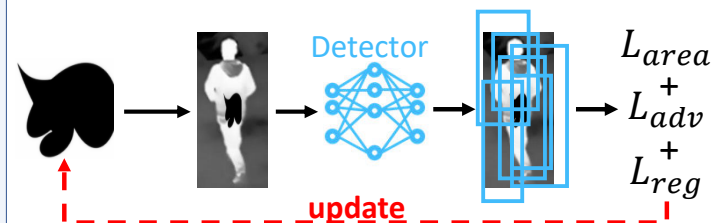
Curve is modeled as Fourier series. Points $(f(t), g(t))$ on boundary satisfy:

$$f(t) + i \cdot g(t) = \sum_{k=-K}^K c_k e^{ikt}, \\ t \in [0, 2\pi]$$

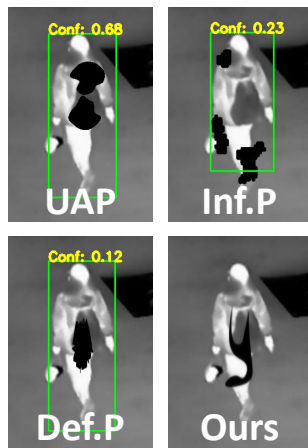
Differentiable Shape-to-Image Mapping



Fourier Shape Patch Optimization



Digital & Physical Experiments



ASR at diverse confidences on different methods

